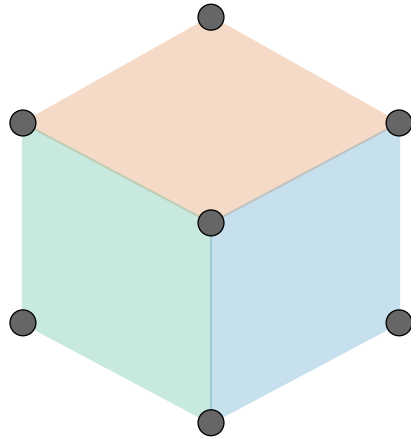


(a) Color code, $d=3$

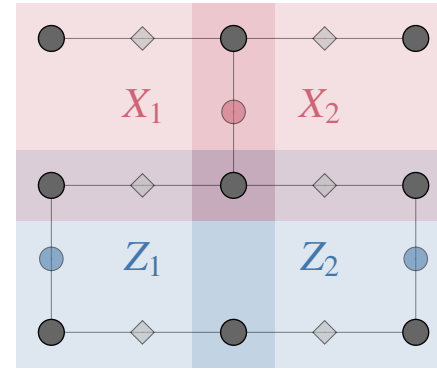
● R ● G ● B faces



7 data qubits + 3 plaquettes (R, G, B)

(b) Heavy-hex surface code, $d=3$

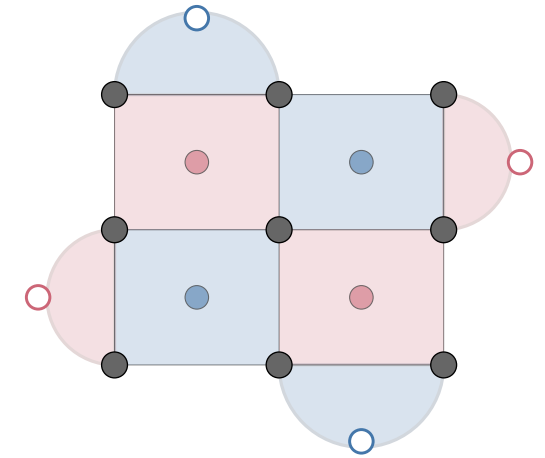
● Z-stab ● X-stab ◆ bridge



9 data + 6 bridges + 3 ancillas

(c) Rotated surface code, $d=3$

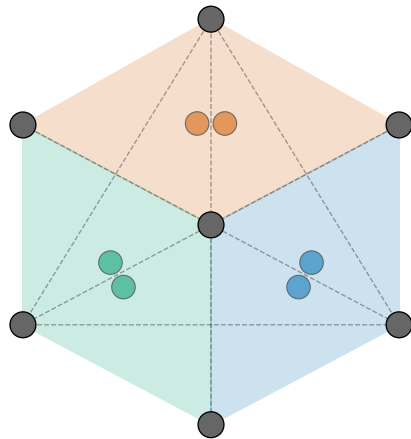
● Z-stab ● X-stab ○ boundary (weight-2)



9 data + 4 bulk ancillas + 4 boundary ancillas

(d) IonQ Forte-1 (color code, $d=3$)

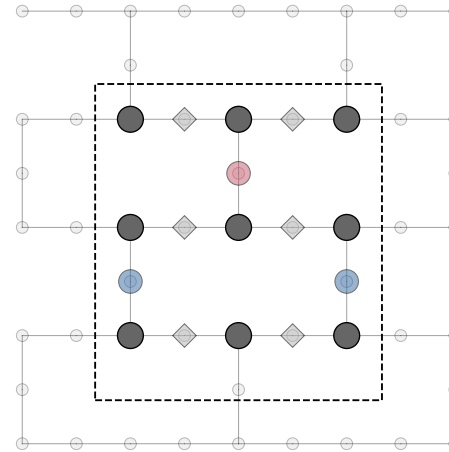
trapped-ion, all-to-all (logical view)



7 data + 6 ancillas (X + Z)

(e) IBM Heron R3, $d=3$

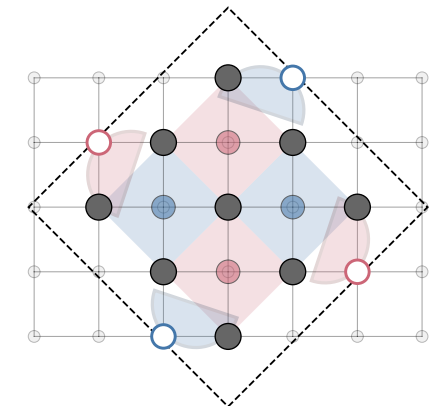
heavy-hex lattice



9 data + 6 bridges + 3 ancillas

(f) IBM Nighthawk, $d=3$

square lattice, 45° rotation



9 data + 4 bulk ancillas + 4 boundary ancillas